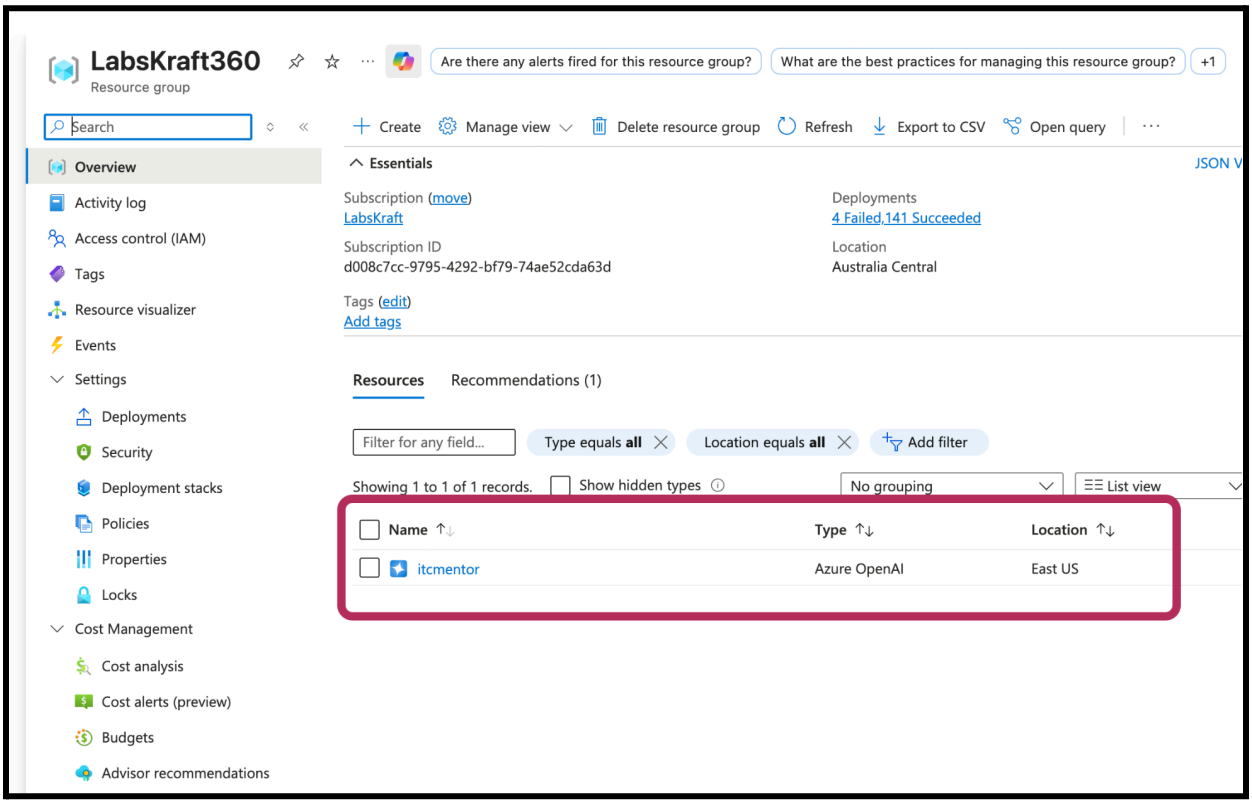
Azure OpenAI Deployment Guide

1. Create Azure OpenAI Resource

- Go to **Azure Portal** → search **OpenAI**.
- Click **Create** → choose **Subscription** and **Resource Group**.
- Provide a **Resource Name** → click **Review + Create**

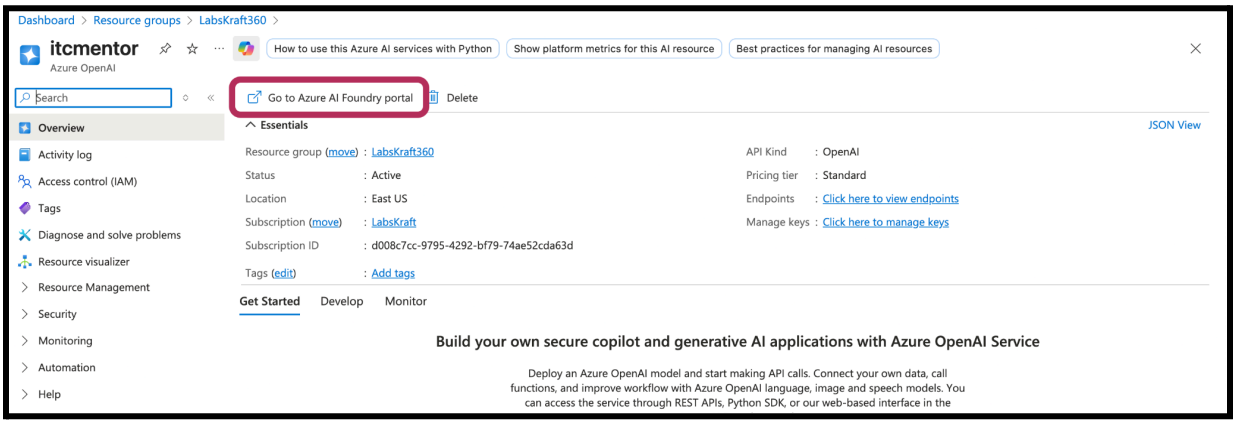


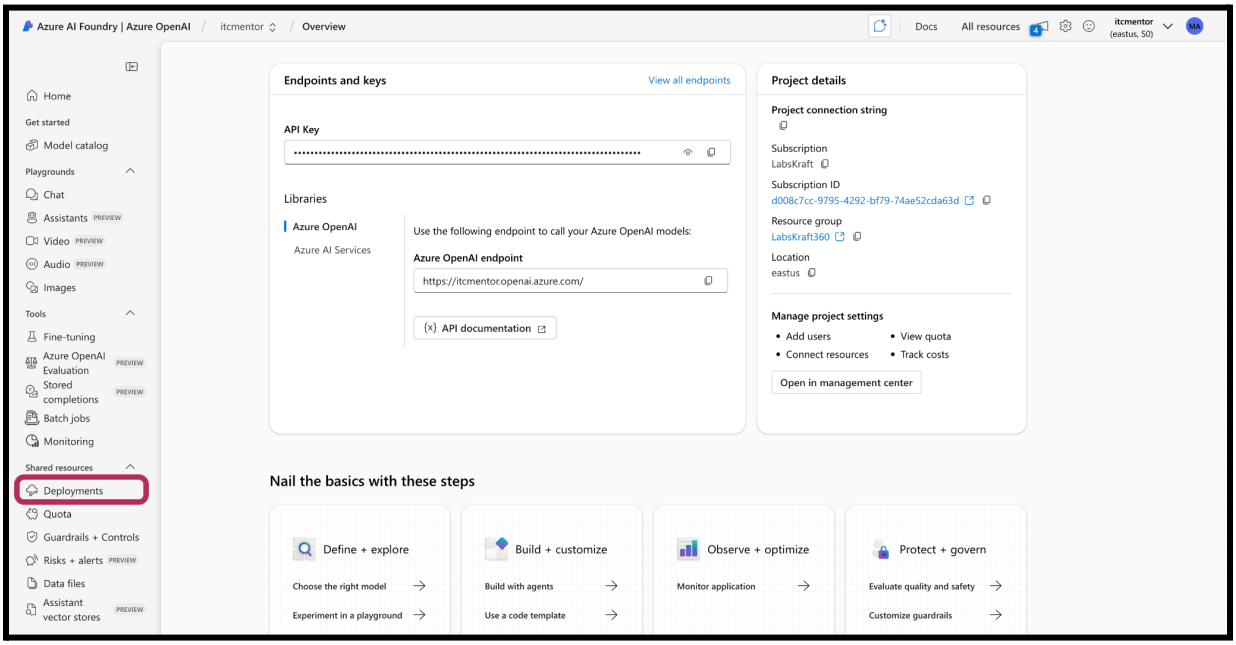
2. Locate the Resource Group

- Go to your **Resource Group**.
- You will see your newly created **Azure OpenAI** resource.

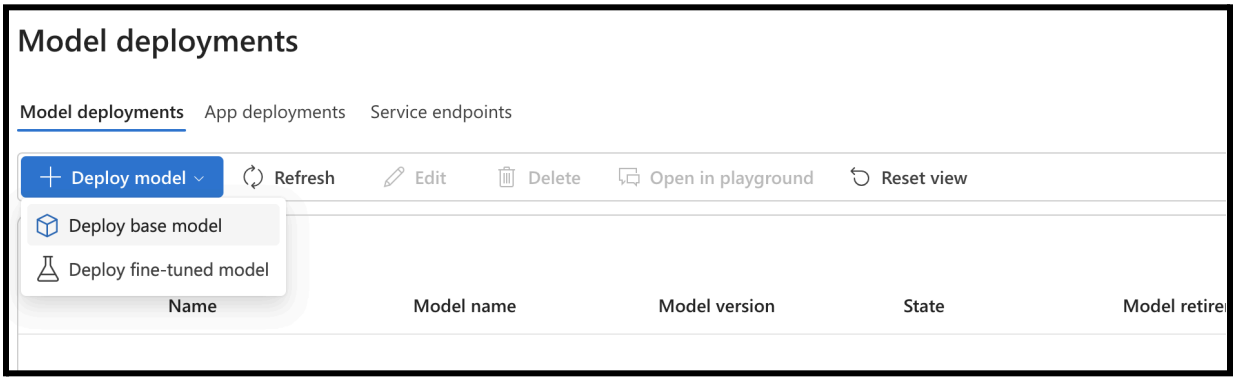
3. Create a Deployment (Model)

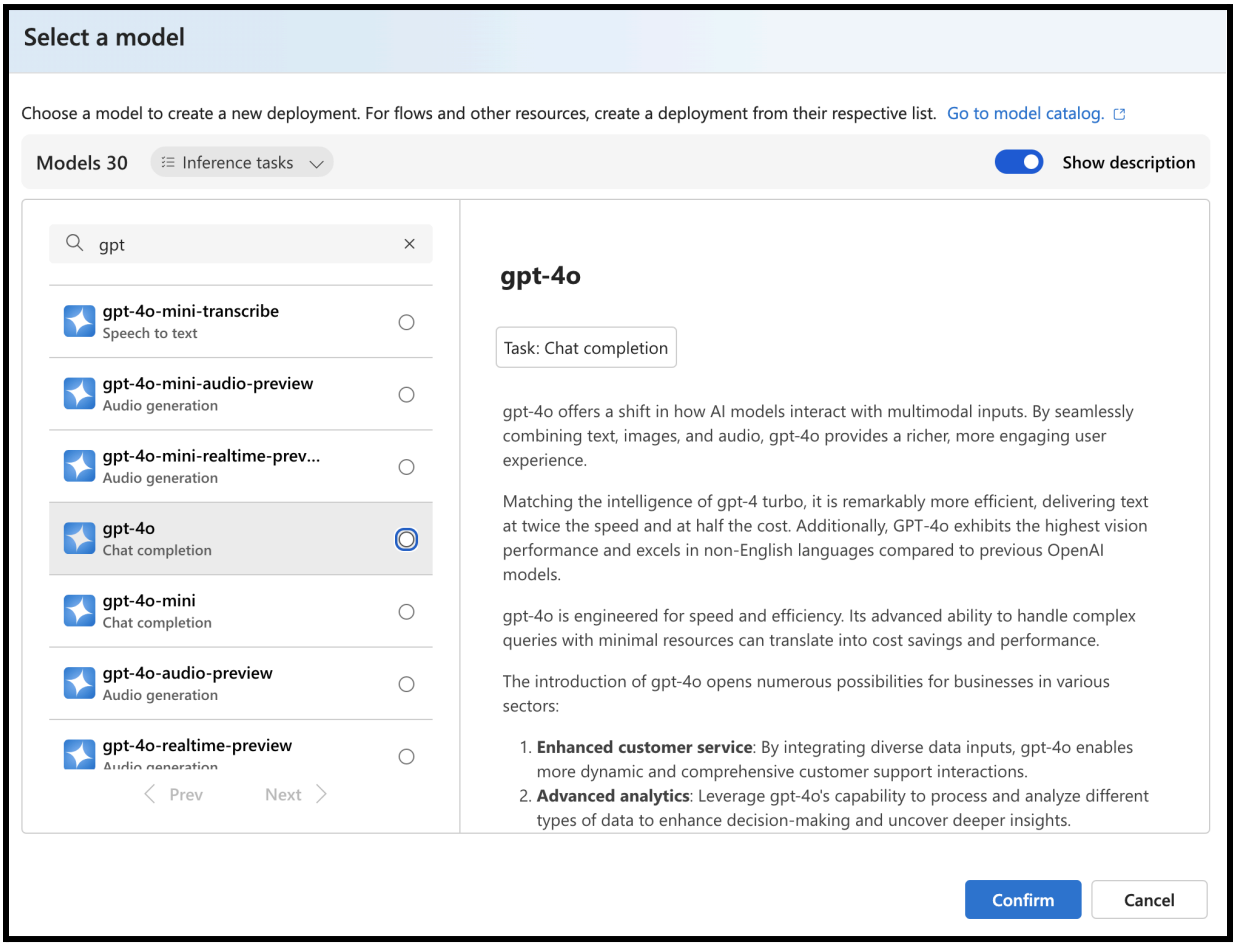
- Click on your **Azure OpenAI resource**.
- Select **Foundry** at the top(shown in screenshot below) → scroll to **Deployments**.



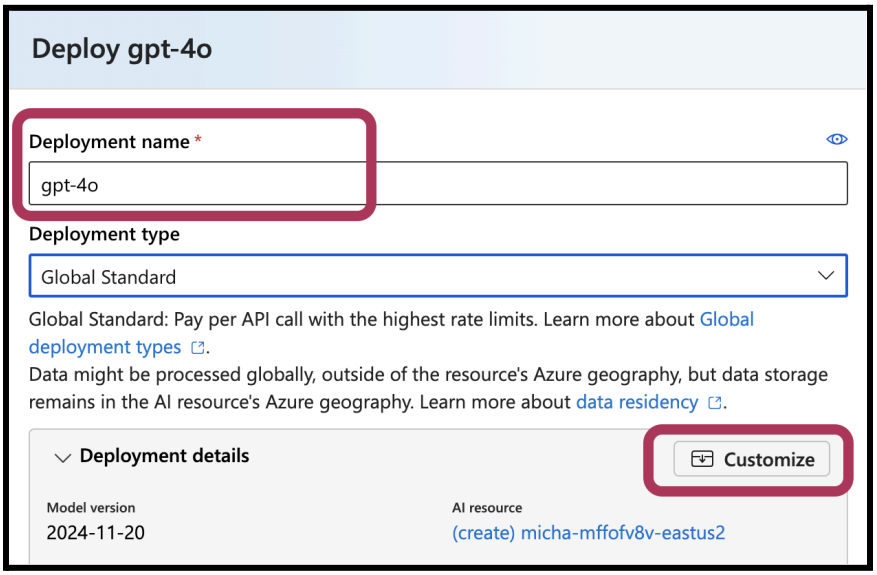


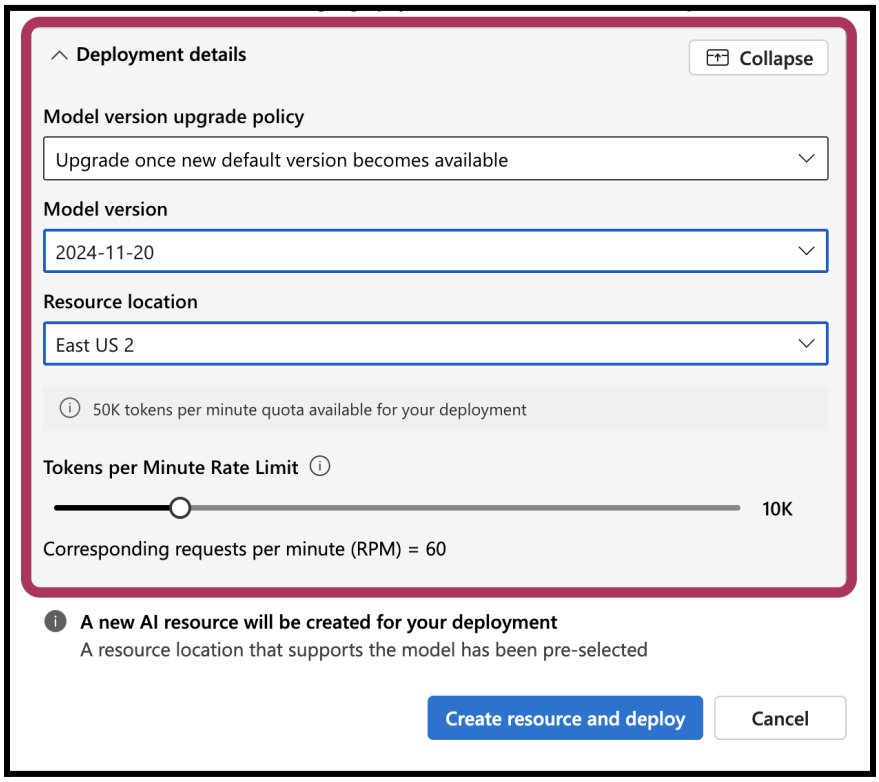
- Click on Deploy model
- Under **Models**, select **gpt-4o**.



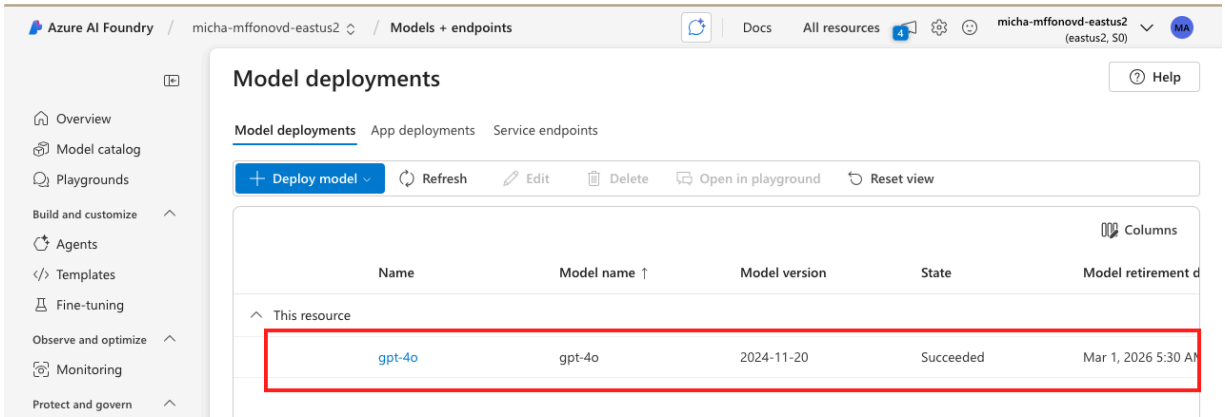


- Configurations for the deployment is shown Below ~



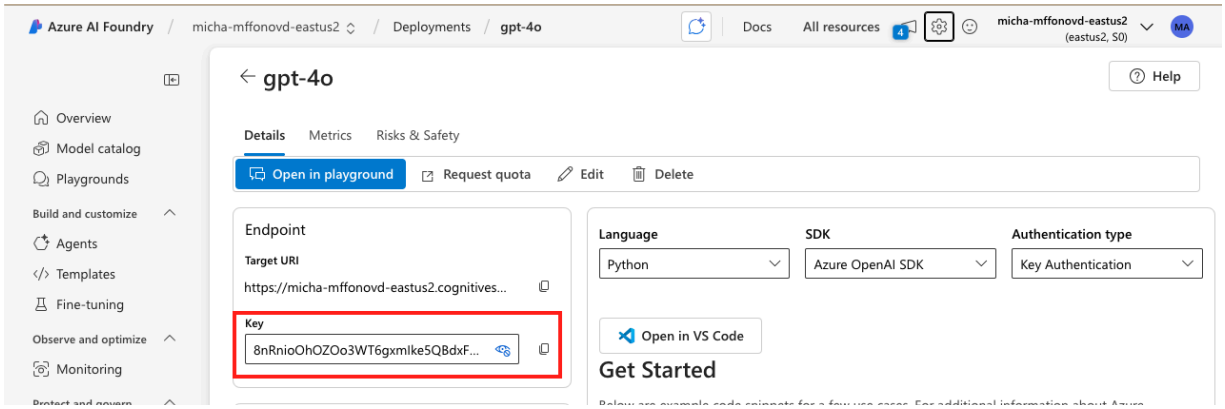


Click **Create Resource and Deploy**
Wait until status shows **Succeeded**.



4. Get Deployment & API Version

- Go back to your **Deployment**
  - Get the api key as shown below



- Scroll down to **Sample Code** → here you'll see:
  - `endpoint`
  - `deployment_name`
  - `api_version` (e.g., `2024-12-01`)

3. Run a basic code sample

This sample demonstrates a basic call to the chat completion API. The call is synchronous.

```
import os
from openai import AzureOpenAI

endpoint = "https://micha-mffonovd-eastus2.cognitiveservices.azure.com/"
model_name = "gpt-4o"
deployment = "gpt-4o"

subscription_key = "<your-api-key>"
api_version = "2024-12-01-preview"

client = AzureOpenAI(
    api_version=api_version,
    azure_endpoint=endpoint,
    api_key=subscription_key,
)

response = client.chat.completions.create(
    messages=[
        {
            "role": "system",
            "content": "You are a helpful assistant.",
        },
        {
            "role": "user",
            "content": "I am going to Paris, what should I see?",
        }
    ],
    max_tokens=4096,
```

5. Setup Development Environment

On Your VM:

1. Open **PyCharm IDE**.
2. Create a new project or use an existing one.
3. Inside the project, create a new Python file:
    - `a1.py` or `chatbot.py`.

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6. Paste the Chatbot Code

```
import os
from dotenv import load_dotenv
from langchain.schema import HumanMessage
from langchain_openai import AzureChatOpenAI

# Load environment variables
load_dotenv()

# Azure OpenAI setup
llm = AzureChatOpenAI(
    api_version=os.getenv("AZURE_OPENAI_API_VERSION"),
    api_key=os.getenv("AZURE_OPENAI_API_KEY"),
    azure_endpoint=os.getenv("AZURE_OPENAI_ENDPOINT"),
    deployment_name=os.getenv("AZURE_OPENAI_DEPLOYMENT_NAME")
)

def chat():
    print(" Azure OpenAI Chatbot (Type 'exit' to quit)\n")
    while True:
        user_input = input("You: ")
        if user_input.lower() == "exit":
            break
        response = llm.invoke([HumanMessage(content=user_input)])
        print(f"Bot: {response.content}\n")

if __name__ == "__main__":
    chat()
```

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7. Create `.env` File

In the same project folder, create a `.env` file and paste your Azure details:

```
AZURE_OPENAI_API_KEY=your_api_key_here
AZURE_OPENAI_ENDPOINT=https://your-resource-name.openai.azure.com
/
AZURE_OPENAI_DEPLOYMENT_NAME=gpt-4o
AZURE_OPENAI_API_VERSION=2024-12-01
```

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8. Install Dependencies

Run these in your terminal (PyCharm or VM):

```
pip install python-dotenv langchain langchain-openai
```

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9. Run the Chatbot

In PyCharm terminal (or VM shell):

```
python chatbot.py
```

You should now see:

```
Azure OpenAI Chatbot (Type 'exit' to quit)

You:
```